

2-rainbow Domination on Cylinders and Tori*

Kung-Jui Pai^{1,+}, Ro-Yu Wu² and Ting-Wei Liu¹

¹ Department of Industrial Engineering and Management, Ming Chi University of Technology, New Taipei City, Taiwan

² Department of Industrial Management, Lunghwa University of Technology, Taoyuan, Taiwan

Abstract

The k -rainbow domination is a variant of the classical domination problem in graphs and is defined as follows. Given an undirected graph $G = (V, E)$, we have a set of k colors and we assign an arbitrary subset of these colors to each vertex of G . If a vertex which is assigned an empty set, then the union of color set of its neighbors must be k colors. This assignment is called the k -rainbow dominating function of G . The minimum sum of numbers of assigned colors over all vertices of G , is called the k -rainbow domination number of G . In this paper, we present some exact values and upper bounds of 2-rainbow domination number on cylinders and tori.

Keywords: Domination; k -rainbow domination; 2-rainbow domination; Cylinder; Torus.

1. Introduction

The k -rainbow domination is a variant of classical domination [7, 8]. Let $G = (V(G), E(G))$ be a finite, simple and undirected graph, where $V(G)$ and $E(G)$ are the vertex and edge sets of G , respectively. For a vertex $v \in V(G)$, the open neighborhood $N(v) = \{u \in V(G) \mid (u, v) \in E(G)\}$. Let $C = \{1, 2, \dots, k\}$ be a set of k colors, and f be a function that assign to each vertex a set of colors chosen from C , that is, $f : V(G) \rightarrow P(\{1, \dots, k\})$. If for each vertex $v \in V(G)$ such that $f(v) = \emptyset$ we have $\bigcup_{u \in N(v)} f(u) = C$ then f is called a k -rainbow dominating function (kRDF) of G . The weight, $\omega(f)$, of a function f is defined as $\omega(f) = \sum_{v \in V(G)} |f(v)|$. The k -rainbow domination number $\gamma_{rk}(G)$ of G is the minimum weight of a k -rainbow dominating function.

Let P_n be the path with n vertices. For example, a 2-rainbow domination of P_5 is shown in Figure 1(a), and $\omega(f) = 3$. We assign a color set $\{1\}$ to V_1, V_5 and assign a color set $\{2\}$ to V_3 . When $f(V_2) = \emptyset$ (respectively, $f(V_4) = \emptyset$), $\bigcup_{u \in N(V_2)} f(u) = \{1, 2\}$ (respectively, $\bigcup_{u \in N(V_4)} f(u) = \{1, 2\}$). For another

example, a 2-rainbow domination of path P_4 is shown in Figure 1(b), and $\omega(f) = 3$. We assign a color set $\{1, 2\}$ to V_2 and assign a color set $\{1\}$ to V_4 . Similarly, the union of color sets of neighbors of V_1 and V_3 are both $\{1, 2\}$. In fact, $\gamma_{r2}(P_5) = 3$ and $\gamma_{r2}(P_4) = 3$ by Observation 3.1 in [4].

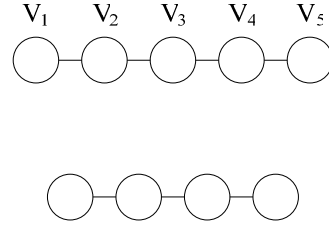


Figure 1: 2-rainbow dominations of P_5 and P_4 . The vertex with a color set $\{1, 2\}$ is labeled by 3, and the vertex with color sets $\{1\}$ and $\{2\}$ are labeled by 1 and 2, respectively.

In [2, 3], Brešar, Henning and Rall introduced the concept of k -rainbow domination of a graph G . The application is that there are k types of guards are used, and vertices v with $f(v) = \emptyset$ must have all types of guards in their neighborhoods. In [4], Brešar and Šumenjak showed that the decision version of 2-rainbow domination of graphs is NP-complete even when restricted to chordal graphs (or bipartite graphs). They also gave the exact values of the 2-rainbow domination numbers for paths, cycles and suns. Thus, further investigations tended to study bounds on k -rainbow domination number for certain families of graphs, such as trees, grids, generalized Petersen graphs and lexicographic product of graphs. This problem is widely studied in [1-6, 9-15].

In [10], we studied 2-rainbow domination on grids. The graphs which we study in this article are cylinders and tori. All grids, cylinders and tori can be defined by means of the Cartesian product of graphs. Let $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ be two graphs which have disjoint vertex sets V_1 and V_2 and disjoint edge sets E_1 and E_2 , respectively. The Cartesian product of G_1 and G_2 is the graph $G_1 \square G_2$ whose set of vertices is the Cartesian product of the sets V_1 and V_2 , so $V(G_1 \square G_2) = V_1 \times V_2$. Two vertices

*This research was partially supported by NSC of R.O.C. under the Grants NSC101-2221-E-131-039 and NSC101-2221-E-262-020.

+Corresponding author: potter@mail.mcut.edu.tw

of $G_1 \square G_2$, say (v_1, v_2) and (u_1, u_2) , are adjacent if and only if $v_1 = u_1$ and $v_2 u_2 \in E(G_2)$, or alternatively, if $v_2 = u_2$ and $v_1 u_1 \in E(G_1)$. The $n \times m$ *grid* can be obtained as the Cartesian product $P_n \square P_m$. Analogously, the $n \times m$ *cylinder* is the Cartesian product $P_n \square C_m$, where C_m stands for the cycle on m vertices. Then, the $n \times m$ *torus* is the Cartesian product of two cycles, say C_n and C_m , respectively.

We recall the definitions of grid, cylinder and torus as follow. For $m \geq n \geq 1$, the grid, $GR_{n,m}$, is a graph with vertex set $V(GR_{n,m}) = \{V_{x,y} \mid 1 \leq x \leq n \text{ and } 1 \leq y \leq m\}$ and two vertices V_{x_1,y_1} and V_{x_2,y_2} are adjacent if and only if $|x_1 - x_2| + |y_1 - y_2| = 1$. The cylinder, $CY_{n,m}$, is a graph with vertex set $V(CY_{n,m}) = \{V_{x,y} \mid 1 \leq x \leq n \text{ and } 1 \leq y \leq m\}$ and two vertices V_{x_1,y_1} and V_{x_2,y_2} are adjacent if and only if $x_1 = x_2$ and $y_1 - y_2 = \pm 1 \pmod{m}$ or $y_1 = y_2$ and $x_1 - x_2 = \pm 1$. For $m \geq n \geq 1$, the torus, $T_{n,m}$, is a graph with vertex set $V(T_{n,m}) = \{V_{x,y} \mid 1 \leq x \leq n \text{ and } 1 \leq y \leq m\}$ and two vertices V_{x_1,y_1} and V_{x_2,y_2} are adjacent if and only if $x_1 = x_2$ and $y_1 - y_2 = \pm 1 \pmod{m}$ or $y_1 = y_2$ and $x_1 - x_2 = \pm 1 \pmod{n}$. For example, Figure 2 depicts $GR_{3,5}$, $CY_{2,3}$ and $T_{3,4}$. In this article, we first present upper bounds of $\gamma_{r2}(CY_{n,m})$, where $2 \leq n \leq 5$ and $m \geq 3$. Then we give exact values of $\gamma_{r2}(T_{3i,j})$, where i is even or j is even. Finally we provide upper bounds of $\gamma_{r2}(T_{n,m})$, where $3 \leq n \leq 5$ and $m \geq n$.

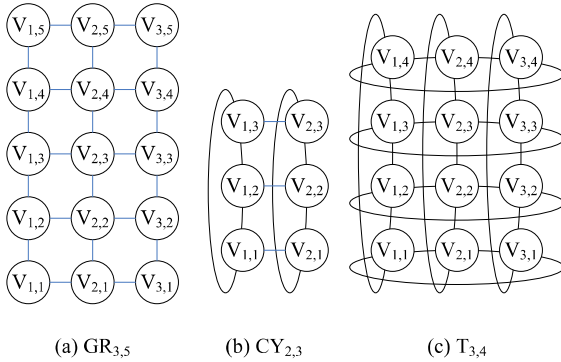


Figure 2: $GR_{3,5}$, $CY_{2,3}$ and $T_{3,4}$.

2. 2-rainbow Domination on Cylinders

In this section, we proceed by giving upper bounds of the 2-rainbow domination number of $CY_{n,m}$, for each $n \in \{2, 3, 4, 5\}$ and $m \geq 3$. For notational convenience, we also refer to a vertex with the color set $\{x\}$ as a vertex with $\{x\}$, and a vertex with an empty set as a vertex with $\emptyset$.

Algorithm P2

Input: A cylinder graph $CY_{2,m}$
 Output: A 2-rainbow dominating set S of $CY_{2,m}$
 for $i \leftarrow 1$ to m do
 if i is odd then
 add $V_{1,i}$ with $\{1\}$ in S ;

 else
 add $V_{2,i}$ with $\{2\}$ in S ;
 end if
end for

Lemma 2.1 $\gamma_{r2}(CY_{2,m}) \leq m$, where $m \geq 3$.

Proof. To achieve this upper bound, we give Algorithm P2 to produce a 2RDF of $CY_{2,m}$. It's easy to verify that $\bigcup_{u \in N(v)} f(u) = \{1, 2\}$ where $f(v) = \emptyset$ and $u, v \in V(CY_{2,m})$. The weight of 2RDF of $CY_{2,m}$ on each row is 1, so $\omega(f) = m$. $\square$

For example, Figures 3(a) and 3(b) show 2-rainbow dominations of $CY_{2,4}$ and $CY_{2,5}$, respectively.

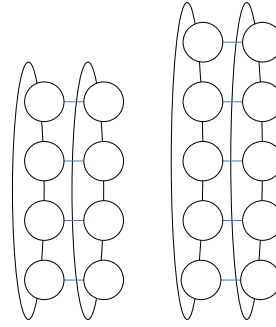


Figure 3: 2-rainbow dominations of $CY_{2,4}$ and $CY_{2,5}$. The vertex with $\{1\}$ and $\{2\}$ are labeled by 1 and 2, respectively.

Algorithm P3A

Input: A cylinder graph $CY_{3,m}$
 Output: A 2-rainbow dominating set S_1 of $CY_{3,m}$
 for $i \leftarrow 1$ to m do
 If $i \equiv 1 \pmod{3}$ then
 add $V_{1,i}$ with $\{1\}$ and $V_{3,i}$ with $\{1\}$ in S_1 ;
 else
 add $V_{2,i}$ with $\{2\}$ in S_1 ;
 end if
end for

Algorithm P3B

Input: A cylinder graph $CY_{3,m}$, which $m \equiv 0 \pmod{8}$
 Output: A 2-rainbow dominating set S_2 of $CY_{3,m}$
 for $i \leftarrow 0$ to $m/8 - 1$ do
 Add $V_{1,8i+1}$ and $V_{3,8i+5}$ with $\{1, 2\}$ in S_2 ;
 Add $V_{1,8i+4}$, $V_{2,8i+7}$, and $V_{3,8i+2}$ with $\{1\}$ in S_2 ;
 Add $V_{1,8i+6}$, $V_{2,8i+3}$, and $V_{3,8i+8}$ with $\{2\}$ in S_2 ;
 end for

Lemma 2.2

$$\gamma_{r2}(CY_{3,m}) \leq \begin{cases} 5m/4, & \text{if } m \equiv 0 \pmod{8}, \\ \lceil 4m/3 \rceil, & \text{otherwise.} \end{cases}$$

Proof. Algorithm P3A produces a 2RDF of $CY_{3,m}$,

and $\omega(f) = \lceil 4m/3 \rceil$. When $m \equiv 0 \pmod{8}$, Algorithm P3B produces another 2RDF of $CY_{3,m}$, and a better upper bound of $\gamma_{r2}(CY_{3,m})$ is $5m/4$. $\square$

For example, Figure 4(a) and 4(b) show 2-rainbow dominations of $CY_{3,6}$ and $CY_{3,8}$, respectively.

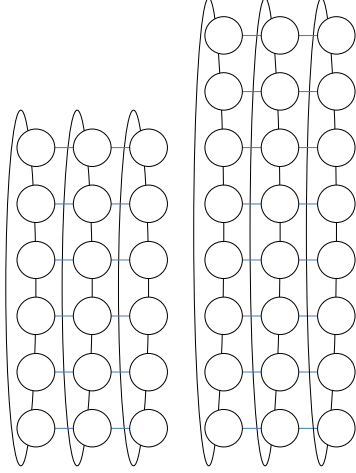


Figure 4: 2-rainbow dominations of $CY_{3,6}$ and $CY_{3,8}$. The vertex with $\{1, 2\}$ is labeled by 3, and the vertex with $\{1\}$ and $\{2\}$ are labeled by 1 and 2, respectively.

Algorithm P4

Input: A cylinder graph $CY_{4,m}$

Output: A 2-rainbow dominating set S of $CY_{3,m}$

for $i \leftarrow 1$ to m do

 if $i \equiv 1 \pmod{3}$ then

 if i is odd then

 add $V_{1,i}$ with $\{1,2\}$ in S ;

 else

 add $V_{4,i}$ with $\{1,2\}$ in S ;

 end if

 end if

 switch ($i \bmod 6$)

 case 1: add $V_{4,i}$ with $\{1\}$ in S ;

 case 2: add $V_{3,i}$ with $\{2\}$ in S ;

 case 3: add $V_{2,i}$ with $\{1\}$ in S ;

 case 4: add $V_{1,i}$ with $\{2\}$ in S ;

 case 5: add $V_{2,i}$ with $\{1\}$ in S ;

 case 0: add $V_{3,i}$ with $\{2\}$ in S ;

 end switch

end for

if $m \equiv 2 \pmod{6}$ then add $V_{m,2}$ with $\{1\}$ in S ;

if $m \equiv 3 \pmod{6}$ then add $V_{m,4}$ with $\{1\}$ in S ;

if $m \equiv 5 \pmod{6}$ then add $V_{m,3}$ with $\{1\}$ in S ;

Lemma 2.3

$$\gamma_{r2}(CY_{4,m}) \leq \begin{cases} \lceil 5m/3 \rceil, & \text{if } m \equiv 0 \pmod{6}, \\ \lceil 5m/3 \rceil + 1, & \text{otherwise.} \end{cases}$$

Proof. Algorithm P4 produces a 2RDF of $CY_{4,m}$. If $m \equiv 0 \pmod{6}$ then the weight of this 2RDF $\omega(f) = \lceil 5m/3 \rceil$ else $\omega(f) = \lceil 5m/3 \rceil + 1$. $\square$

For example, Figures 5(a), 5(b), 5(c), 5(d), 5(e), and 5(f) show 2-rainbow dominations of $CY_{4,3}$, $CY_{4,4}$, $CY_{4,5}$, $CY_{4,6}$, $CY_{4,7}$ and $CY_{4,8}$, respectively.

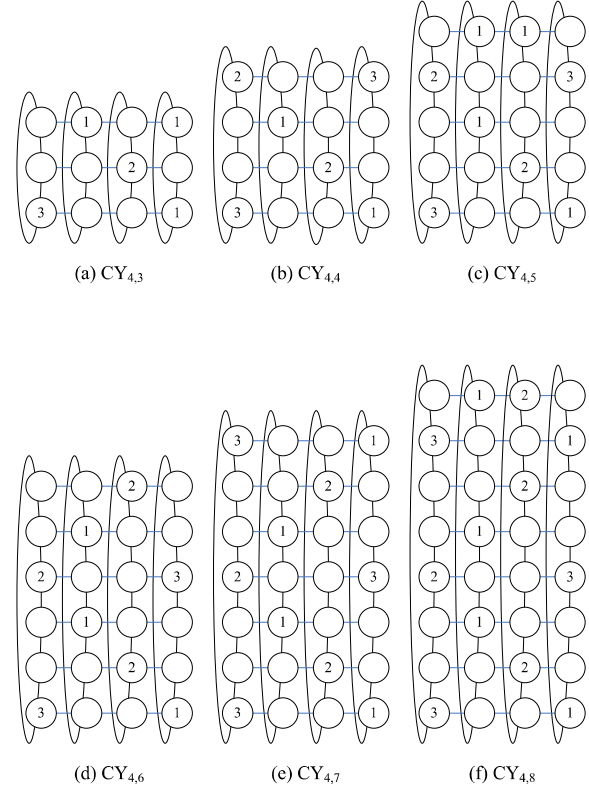


Figure 5: 2-rainbow dominations of $CY_{4,3}$, $CY_{4,4}$, $CY_{4,5}$, $CY_{4,6}$, $CY_{4,7}$ and $CY_{4,8}$. The vertex with $\{1, 2\}$ is labeled by 3, and the vertex with $\{1\}$ and $\{2\}$ are labeled by 1 and 2, respectively.

Lemma 2.4 $\gamma_{r2}(CY_{5,m}) \leq 2m + 1$, where $m \in \{3, 4, 6, 7, 11, 12\}$

Proof. To achieve this upper bound, we directly give 2RDFs by graph drawing as shown in Figures 6(a), 6(b), 6(c), 6(d), 6(e), and 6(f). $\square$

We recall the 2RDFs of $GR_{5,5}$ in [10]. Let Pattern A_5 be a 2-rainbow domination of $GR_{5,5}$ which is consist of $V_{3,3}$ with $\{1, 2\}$, vertices $V_{1,2}$, $V_{2,5}$, $V_{4,1}$, $V_{5,4}$ with $\{1\}$ and vertices $V_{1,4}$, $V_{2,1}$, $V_{4,5}$, $V_{5,2}$ with $\{2\}$. Let Pattern B_5 be another 2-rainbow domination of $GR_{5,5}$ which is consist of $V_{3,3}$ with $\{1, 2\}$, vertices $V_{1,2}$, $V_{2,5}$, $V_{4,1}$, $V_{5,4}$ with $\{2\}$ and vertices $V_{1,4}$, $V_{2,1}$, $V_{4,5}$, $V_{5,2}$ with $\{1\}$. For preparation of next lemma, we give another 2 patterns. Pattern B_3 is obtained from Pattern B_5 by removing first and last rows, and Pattern B_4 is obtained from Pattern B_5 by removing first row. Figures 7(a), 7(b), 7(c) and 7(d) shows

Pattern A_5 , B_5 , B_3 and B_4 , respectively.

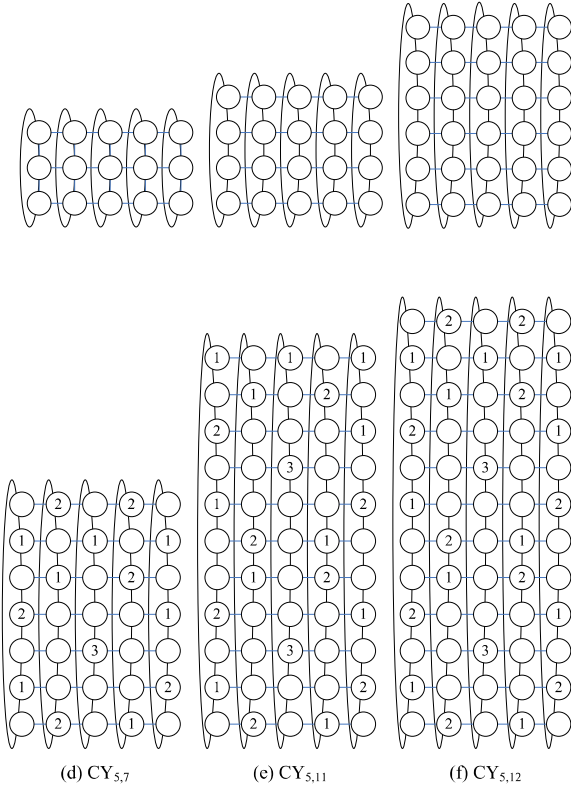


Figure 6: 2-rainbow dominations of $CY_{5,3}$, $CY_{5,4}$, $CY_{5,6}$, $CY_{5,7}$, $CY_{5,11}$ and $CY_{5,12}$. The vertex with $\{1, 2\}$ is labeled by 3, and the vertex with $\{1\}$ and $\{2\}$ are labeled by 1 and 2, respectively.

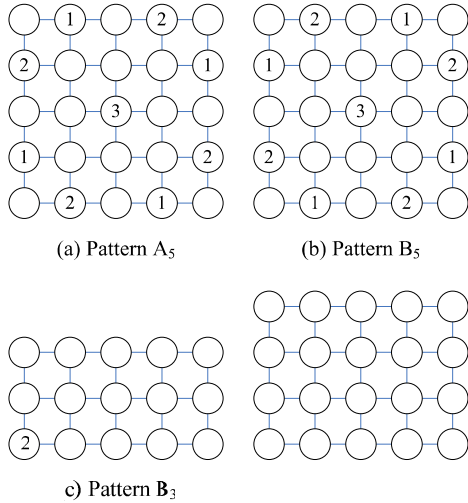


Figure 7: Patterns A_5 , B_5 , B_3 and B_4 . The vertex with $\{1, 2\}$ is labeled by 3, and the vertex with $\{1\}$ and $\{2\}$ are labeled by 1 and 2, respectively.

Then, the concatenation of Pattern X and Pattern Y, written $(X) \circ (Y)$, is Pattern X overlaid by Pattern Y. Let $k(X)$ be $\underbrace{(X) \circ \dots \circ (X)}_k$. For example, Figure 8 show $(A_5) \circ (B_3)$.

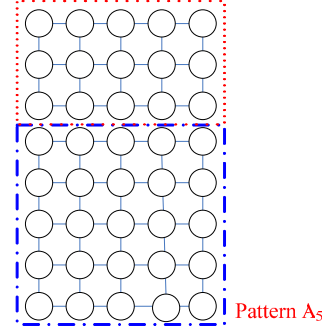


Figure 8: $(A_5) \circ (B_3)$.

Lemma 2.5 $\gamma_{r2}(CY_{5,m}) \leq 2m$, where $m \geq 5$ and $m \notin \{6, 7, 11, 12\}$.

Proof. Case 1: $\gamma_{r2}(CY_{5,5}) \leq 10$. $CY_{5,5}$ can be obtained from $GR_{5,5}$ by adding edges $(V_{i,1}, V_{i,5})$ where $1 \leq i \leq 5$. Obviously Pattern A_5 can produce a 2RDF of $CY_{5,5}$.

Case 2: $\gamma_{r2}(CY_{5,8}) \leq 16$. To achieve this upper bound, we give a 2RDF, $(A_5) \circ (B_3)$, of $CY_{5,8}$. We note that there exist edges $(V_{i,1}, V_{i,8})$ in $CY_{5,8}$, where $1 \leq i \leq 5$. Pattern B_5 can be composed from last 4 rows and first row of $CY_{5,8}$.

Case 3: $\gamma_{r2}(CY_{5,9}) \leq 18$. Similar to case 2, we give a 2RDF, $(A_5) \circ (B_4)$, of $CY_{5,9}$.

Case 4: $\gamma_{r2}(CY_{5,m}) \leq 2m$, where $m \geq 10$ and $m \equiv 0 \pmod{5}$. To achieve this upper bound, we give a 2RDF, $k(A_5)$, of $CY_{5,m}$, where $k = m/5$.

Case 5: $\gamma_{r2}(CY_{5,m}) \leq 2m$, where $m \geq 13$ and $m \pmod{5} \in \{3, 4\}$. $k(A_5) \circ (B_j)$ can produce a 2RDF of $CY_{5,m}$, where $j = m \pmod{5}$ and $k = \lfloor m/5 \rfloor$.

Case 6: $\gamma_{r2}(CY_{5,m}) \leq 2m$, where $m \geq 16$ and $m \pmod{5} \in \{1, 2\}$. To achieve this upper bound, we give a 2RDF, $k(A_5) \circ (B_3) \circ (A_5) \circ (B_{2+j})$, of $CY_{5,m}$, where $j = m \pmod{5}$ and $k = \lfloor m/5 \rfloor - 2$.

In all cases, we note that the weight of 2RDF on each row is 2, so the upper bound of $\gamma_{r2}(CY_{5,m})$ is $2m$, where $m \geq 5$ and $m \notin \{6, 7, 11, 12\}$. $\square$

We summarize Lemmas 2.1 to 2.5 as follows.

Theorem 2.6 Some upper bounds of $\gamma_{r2}(CY_{n,m})$ are shown as below, which $m \geq 3$.

- (1) $\gamma_{r2}(CY_{2,m}) \leq m$,
- (2) $\gamma_{r2}(CY_{3,m}) \leq \begin{cases} 5m/4, & \text{if } m \equiv 0 \pmod{8}, \\ \lceil 4m/3 \rceil, & \text{otherwise.} \end{cases}$
- (3) $\gamma_{r2}(CY_{4,m}) \leq \begin{cases} \lceil 5m/3 \rceil, & \text{if } m \equiv 0 \pmod{6}, \\ \lceil 5m/3 \rceil + 1, & \text{otherwise.} \end{cases}$
- (4) $\gamma_{r2}(CY_{5,m}) \leq \begin{cases} 2m+1, & m \in \{3, 4, 6, 7, 11, 12\} \\ 2m, & \text{otherwise.} \end{cases}$

As described above, we give 2RDFs to provide the upper bounds of $CY_{n,m}$, where $2 \leq n \leq 5$. In order to know the difference between lower bounds of

$\gamma_{r2}(CY_{n,m})$ and upper bounds of $\gamma_{r2}(CY_{n,m})$. With the help of computer programs, we have computed $\gamma_{r2}(CY_{n,m})$ by using brute-force algorithms. These algorithms test all feasible solutions in $CY_{n,m}$ to find a minimum weight of 2RDF, denoted by γ_{r2}^P . Table 1 shows the results of brute-force algorithms and the weight of our 2RDFs for some smaller n and m . From this table, we observe that the upper bounds of $\gamma_{r2}(CY_{n,m})$ by Theorem 2.6 completely match γ_{r2}^P for those smaller n and m .

Table 1: Comparison of γ_{r2}^P and the upper bounds of $\gamma_{r2}(CY_{n,m})$ by Theorem 2.6

Graph	γ_{r2}^P	UB ⁺
$CY_{2,m}, 3 \leq m \leq 13$	m	m
$CY_{3,3}$	4	4
$CY_{3,4}$	6	6
$CY_{3,5}$	7	7
$CY_{3,6}$	8	8
$CY_{3,7}$	10	10
$CY_{3,8}$	10	10
$CY_{3,9}$	12	12
$CY_{4,3}$	6	6
$CY_{4,4}$	8	8
$CY_{4,5}$	10	10
$CY_{4,6}$	10	10
$CY_{4,7}$	13	13
$CY_{5,3}$	7	7
$CY_{5,4}$	9	9
$CY_{5,5}$	10	10
$CY_{5,6}$	13	13
$CY_{6,3}$	8	
$CY_{6,4}$	11	
$CY_{6,4}$	13	

⁺ UB: the upper bounds of $\gamma_{r2}(CY_{n,m})$ by Theorem 2.6

3. 2-rainbow Domination on Tori

Since $T_{n,m} \cong T_{m,n}$, we only discuss upper bounds of $\gamma_{r2}(T_{n,m})$ where $3 \leq n \leq m$.

Algorithm C3

Input: A torus graph $T_{3,m}$

Output: A 2-rainbow dominating set S of $T_{3,m}$
for $i \leftarrow 1$ to m do

 add $V_{(3-i \bmod 3),i}$ with $\{2-i \bmod 2\}$ in S ;
end for

if $m \bmod 6 \neq 0$ then add $V_{3,m}$ with $\{2\}$ in S ;

if $m \equiv 4 \pmod{6}$ then add $V_{1,m}$ with $\{1\}$ in S ;

Lemma 3.1 $\gamma_{r2}(T_{3,m}) \leq \begin{cases} m, & \text{if } m \equiv 0 \pmod{6}, \\ m+2, & \text{if } m \equiv 4 \pmod{6}, \\ m+1, & \text{otherwise.} \end{cases}$

Proof. To achieve this upper bound, we give Algorithm C3 to produce a 2RDF of $T_{3,m}$. $\square$

For example, Figures 9(a), 9(b), 9(c), 9(d), 9(e), and 9(f) show 2-rainbow dominations of $T_{3,3}$, $T_{3,4}$, $T_{3,5}$, $T_{3,6}$, $T_{3,7}$ and $T_{3,8}$, respectively.

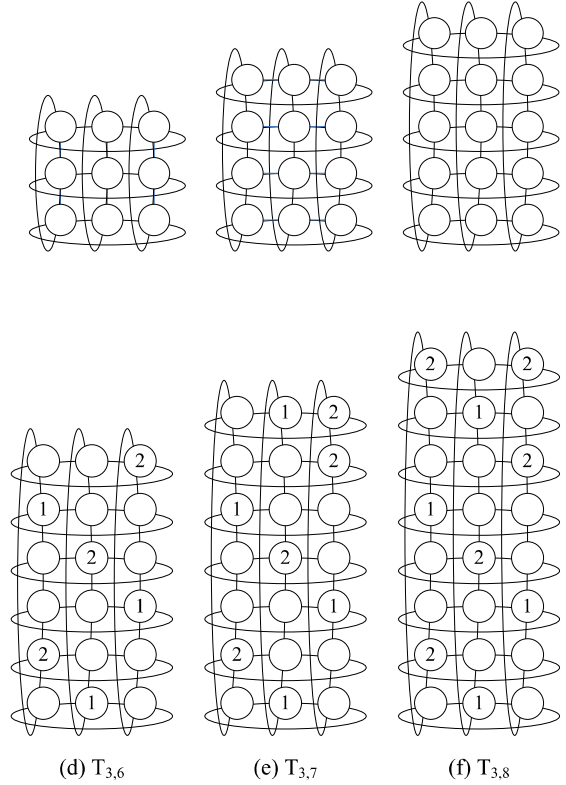


Figure 9: 2-rainbow dominations of $T_{3,3}$, $T_{3,4}$, $T_{3,5}$, $T_{3,6}$, $T_{3,7}$ and $T_{3,8}$. The vertex with $\{1, 2\}$ is labeled by 3, and the vertex with $\{1\}$ and $\{2\}$ are labeled by 1 and 2, respectively.

Lemma 3.2 For a 4-regular graph G , $|V(G)| / \gamma_{r2}(G) \leq 3$.

Proof. For 2-rainbow domination, a vertex with $\emptyset$ is dominated by either a vertex with $\{1, 2\}$ or a vertex u with $\{1\}$ and a vertex v with $\{2\}$.

Case 1: A vertex with $\emptyset$ is dominated by a vertex with $\{1, 2\}$. For 4-regular graph G , a vertex v with $\{1, 2\}$ at most dominates 4 vertex with $\emptyset$. Let H be the induced subgraph of G by the vertex set $N[v]$ where $N[v] = N(v) \cup v$. $|V(H)| / \gamma_{r2}(H) = 5 / 2 = 2.5 \leq 3$.

Case 2: A vertex with $\emptyset$ is dominated by a vertex u with $\{1\}$ and a vertex v with $\{2\}$. For 4-regular graph G , a vertex v with $\{1\}$ (or $\{2\}$) at most adjacent to 4 vertex with $\emptyset$. If there are $4k$ vertices with $\emptyset$ in G , then there are at least k vertices with $\{1\}$ and k vertices with $\{2\}$. In this graph G , $|V(G)| / \gamma_{r2}(G) = (4k + k + k) / (k + k) = 3$. $\square$

By Lemma 3.2, it directly derives the following corollary.

Corollary 3.3 For a 4-regular graph G , if there exist a 2RDF which $|V(G)| / \omega(f) = 3$, then $\gamma_{r2}(G) = \omega(f)$.

For example, Figure 9(d) show a 2RDF of $T_{3,6}$. We can derive $\gamma_{r2}(T_{3,6}) = 6$ by Corollary 3.3.

Theorem 3.4 $\gamma_{r2}(T_{3i,3j}) = 3ij$, where $i \leq j$ and i is even or j is even.

Proof. Figure 9(d) show a 2RDF, denoted by R_{36} , of $T_{3,6}$. Since $T_{6,3} \cong T_{3,6}$, we let a 2RDF, denote by R_{63} , of $T_{6,3}$ be obtained by swapping the coordinates of x and y in R_{36} .

Case 1: When j is even. Let $k_1 = 3i / 3$ and $k_2 = 3j / 6$. A 2RDF of $T_{3i,3j}$ can be composed by $k_1 \times k_2$ R_{36} .

Case 2: When i is even. Let $k_1 = 3i / 6$ and $k_2 = 3j / 3$. A 2RDF of $T_{3i,3j}$ can be composed by $k_1 \times k_2$ R_{63} .

When both i and j are even, a 2RDF of $T_{3i,3j}$ can be composed by either R_{36} or R_{63} . Then, $T_{3i,3j}$ is a 4-regular graph and $|V(G)| / \omega(f) = 3i \times 3j / 3ij = 3$. By Corollary 3.3, $\gamma_{r2}(T_{3i,3j}) = 3ij$. $\square$

For example, Figure 10 show that a 2RDF of $T_{6,12}$ can be composed by 2×2 R_{63} .

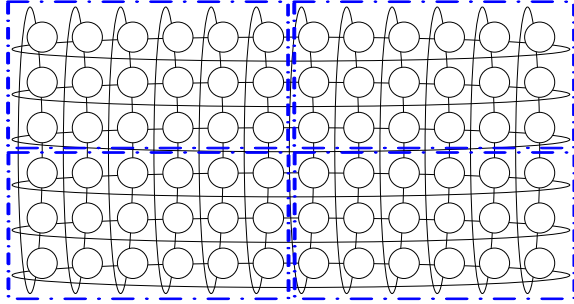


Figure 10: A 2RDF of $T_{6,12}$. The vertex with $\{1\}$ and $\{2\}$ are labeled by 1 and 2, respectively.

Lemma 3.5 $\gamma_{r2}(T_{4,3}) \leq 6$, $\gamma_{r2}(T_{4,4}) \leq 7$, $\gamma_{r2}(T_{4,5}) \leq 8$, $\gamma_{r2}(T_{4,6}) \leq 10$, $\gamma_{r2}(T_{4,7}) \leq 12$, $\gamma_{r2}(T_{4,8}) \leq 12$, $\gamma_{r2}(T_{4,9}) \leq 15$ and $\gamma_{r2}(T_{4,10}) \leq 16$.

Proof. To achieve this upper bound, we directly give 2RDFs of $T_{4,m}$ by graph drawing for $3 \leq m \leq 8$. Figures 11(a), 11(b), 11(c), 11(d), 11(e), and 11(f) show 2-rainbow dominations of $T_{4,3}$, $T_{4,4}$, $T_{4,5}$, $T_{4,6}$, $T_{4,7}$ and $T_{4,8}$, respectively. Let R_{44} and R_{45} be the 2RDF of $T_{4,4}$ and $T_{4,5}$ as shown in Figure 11, respectively. Since 2RDFs of first row of R_{44} and R_{45} are the same, and 2RDFs of last row of R_{44} and R_{45} are the same. A 2RDF of $T_{4,9}$ (respectively, $T_{4,10}$) can be composed by $(R_{44}) \circ (R_{45})$ (respectively, $(R_{45}) \circ (R_{44})$). $\square$

To conveniently read the results of Lemma 3.5, we list them in Table 2.

Table 2: $\omega(f)$ of $T_{4,m}$, where $3 \leq m \leq 10$

m	3	4	5	6	7	8	9	10
$\omega(f)$ of $T_{4,m}$	6	7	8	10	12	12	15	16

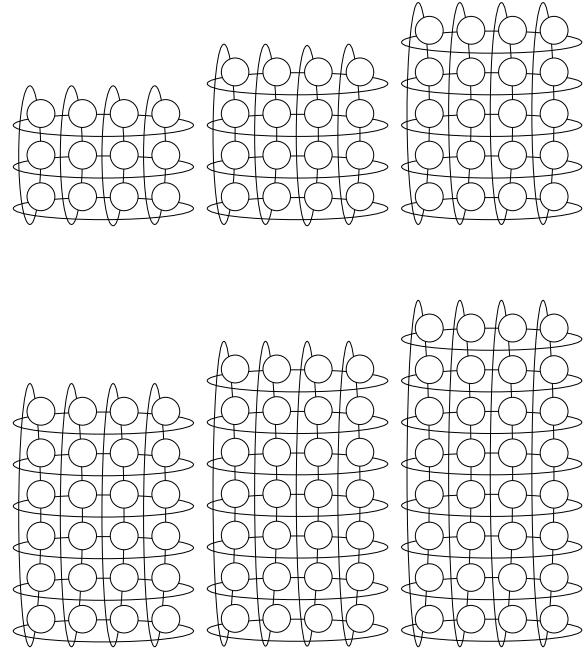


Figure 11: 2-rainbow dominations of $T_{4,3}$, $T_{4,4}$, $T_{4,5}$, $T_{4,6}$, $T_{4,7}$ and $T_{4,8}$. The vertex with $\{1, 2\}$ is labeled by 3, and the vertex with $\{1\}$ and $\{2\}$ are labeled by 1 and 2, respectively.

Lemma 3.6 Let $m \geq 11$, $j = m \bmod 8$ and $k = \lfloor m/8 \rfloor$,

$$\gamma_{r2}(T_{4,m}) \leq \begin{cases} \omega(f) \text{ of } k(R_{48}), & \text{if } j=0, \\ \omega(f) \text{ of } (k-1)(R_{48}) \circ (R_{45}) \circ (R_{44}), & \text{if } j=1, \\ \omega(f) \text{ of } (k-1)(R_{48}) \circ 2(R_{45}), & \text{if } j=2, \\ \omega(f) \text{ of } (k-1)(R_{48}) \circ 2(R_{45}) \circ (R_{44}), & \text{if } j=6, \\ \omega(f) \text{ of } k(R_{48}) \circ (R_{4j}), & \text{otherwise.} \end{cases}$$

Proof. Let R_{43} , R_{44} , R_{45} , R_{47} , and R_{48} be the 2RDF of $T_{4,3}$, $T_{4,4}$, $T_{4,5}$, $T_{4,7}$ and $T_{4,8}$ as shown in Figure 11, respectively. Since 2RDFs of first row of R_{43} , R_{44} , R_{45} , R_{47} , and R_{48} are the same and 2RDFs of last row of R_{43} , R_{44} , R_{45} , R_{47} , and R_{48} are the same. A 2RDF of $T_{4,m}$ can be composed by R_{43} , R_{44} , R_{45} , R_{47} , and R_{48} . The upper bound of $\gamma_{r2}(T_{4,m})$ can be directly derived. $\square$

From Lemmas 3.5 and 3.6, it directly derives the following corollary.

Corollary 3.7 $\gamma_{r2}(T_{4,m}) \leq \lceil 8m/5 \rceil$, where $m \geq 4$.

Algorithm C5

Input: A torus graph $T_{5,m}$

Output: A 2-rainbow dominating set S of $T_{5,m}$

if $m \equiv 1 \pmod{3}$ then

add $V_{1,m-3}$, $V_{1,m-2}$, $V_{3,m}$ and $V_{4,m-2}$ with $\{1\}$ in S ;

add $V_{2,m-1}$, $V_{3,m-3}$, $V_{5,m-1}$ and $V_{5,m}$ with $\{2\}$ in S ;

```

end if
if  $m \equiv 2 \pmod{3}$  then
  add  $V_{1,m-4}, V_{2,m-2}, V_{3,m}, V_{4,m-3}$  and  $V_{5,m-1}$  with  $\{1\}$ 
  in  $S$ ;
  add  $V_{1,m-3}, V_{2,m-1}, V_{3,m-4}, V_{4,m-2}$  and  $V_{5,m}$  with  $\{2\}$ 
  in  $S$ ;
end if
 $t \leftarrow \lfloor m/3 \rfloor - 1$ ;
if  $i \equiv 0 \pmod{3}$  then  $t \leftarrow t + 1$ ;
for  $i \leftarrow 1$  to  $t$  do
  add  $V_{1,3i-2}, V_{4,3i-1}$  and  $V_{3,3i}$  with  $\{1\}$  in  $S$ ;
  add  $V_{3,3i-2}, V_{2,3i-1}$  and  $V_{5,3i}$  with  $\{2\}$  in  $S$ ;
end for

```

Lemma 3.8 $\gamma_{r2}(T_{5,m}) \leq 2m$, where $m \geq 5$.

Proof. Algorithm C5 produces 2RDFs of $T_{5,m}$ recursively. Let R_{53} , R_{54} and R_{55} be the 2RDFs of $T_{5,3}$, $T_{5,4}$ and $T_{5,5}$ by Algorithm C5, respectively. If $m \equiv 0 \pmod{3}$, $(\lfloor m/3 \rfloor)(R_{53})$ is a 2RDF of $T_{5,m}$. If $m \equiv 1 \pmod{3}$, $(\lfloor m/3 \rfloor - 1)(R_{53}) \circ (R_{54})$ is a 2RDF of $T_{5,m}$. If $m \equiv 2 \pmod{3}$, $(\lfloor m/3 \rfloor - 1)(R_{53}) \circ (R_{55})$ is a 2RDF of $T_{5,m}$. Since 2RDFs of first row of R_{53} , R_{54} and R_{55} are the same, and 2RDFs of last row of R_{53} , R_{54} and R_{55} are the same, a 2RDF of $T_{5,m}$ can be composed by R_{53} , R_{54} and R_{55} . Then $\omega(f)$ of each row is 2 by Algorithm C, the upper bound of $\gamma_{r2}(T_{5,m})$ is $2m$. $\square$

For example, Figures 12(a), 12(b) and 12(c) show 2-rainbow dominations of $T_{5,3}$, $T_{5,4}$ and $T_{5,5}$, respectively.

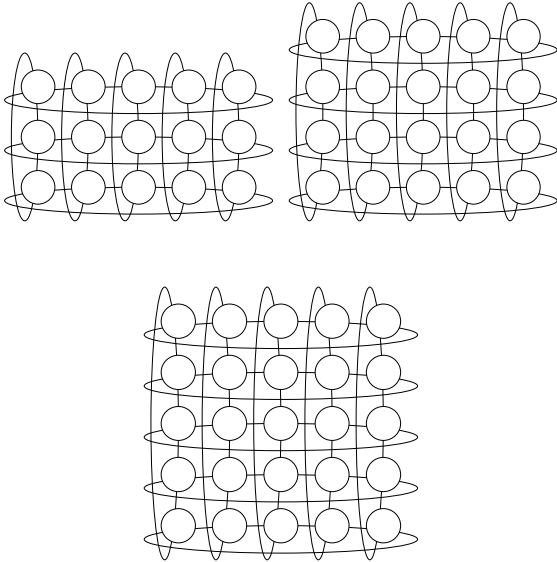


Figure 12: 2-rainbow dominations of $T_{5,3}$, $T_{5,4}$ and $T_{5,5}$. The vertex with $\{1\}$ and $\{2\}$ are labeled by 1 and 2, respectively.

We summarize Lemmas 3.1, 3.8 and Corollary 3.7 as follows.

Theorem 3.9 Some upper bounds of $\gamma_{r2}(T_{n,m})$ are shown as below, which $3 \leq n \leq m$.

- (1) $\gamma_{r2}(T_{3,m}) \leq \begin{cases} m, & \text{if } m \equiv 0 \pmod{6}, \\ m+2, & \text{if } m \equiv 4 \pmod{6}, \\ m+1, & \text{otherwise,} \end{cases}$
- (2) $\gamma_{r2}(T_{4,m}) \leq \lceil 8m/5 \rceil$,
- (3) $\gamma_{r2}(T_{5,m}) \leq 2m$.

As described above, we give 2RDFs to provide the upper bounds of $T_{n,m}$, where $3 \leq n \leq 5$. In order to know the difference between lower bounds of $\gamma_{r2}(T_{n,m})$ and upper bounds of $\gamma_{r2}(T_{n,m})$. With the help of computer programs, we have computed $\gamma_{r2}(T_{n,m})$ by using brute-force algorithms. These algorithms test all feasible solutions in $T_{n,m}$ to find a minimum weight of 2RDF, denoted by γ_{r2}^P . Table 3 shows the results of brute-force algorithms and the weight of our 2RDFs for some smaller n and m . From this table, we observe that the upper bounds of $\gamma_{r2}(T_{n,m})$ by Theorem 3.9 completely match γ_{r2}^P for those smaller n and m .

Table 3: Comparison of $\gamma_{r2}^P(T_{n,m})$ and the upper bounds of $\gamma_{r2}(T_{n,m})$ by Theorem 3.9

Graph	γ_{r2}^P	UB ⁺
$T_{3,3}$	4	4
$T_{3,4}$	6	6
$T_{3,5}$	6	6
$T_{3,6}$	6	6
$T_{3,7}$	8	8
$T_{3,8}$	9	9
$T_{3,9}$	10	10
$T_{3,10}$	12	12
$T_{4,4}$	7	7
$T_{4,5}$	8	8
$T_{4,6}$	10	10
$T_{4,7}$	12	12
$T_{4,8}$	12	12
$T_{5,5}$	10	10
$T_{5,6}$	12	12
$T_{6,6}$	12	12

⁺ UB: the upper bounds of $\gamma_{r2}(T_{n,m})$ by Theorem 3.9

4. Conclusion

In this paper, we study 2-rainbow domination on cylinders and tori. Then, we design some algorithms and 2RDFs to provide upper bounds of $\gamma_{r2}(CY_{n,m})$ and $\gamma_{r2}(T_{n,m})$, and a exact value of $\gamma_{r2}(T_{3i,3j})$ where i is even or j is even.

With the help of computer programs, we have computed $\gamma_{r2}^P(CY_{n,m})$ and $\gamma_{r2}^P(T_{n,m})$ by using brute-force algorithms. The execution results show that the upper bounds of $\gamma_{r2}(CY_{n,m})$ by Theorem 2.6 (respectively, the upper bounds of $\gamma_{r2}(T_{n,m})$ by Theorem 3.9) completely match $\gamma_{r2}^P(CY_{n,m})$

(respectively, $\gamma_{r2}^P(T_{n,m})$) for those smaller n and m .

Since 2RDFs of $CY_{n_1,m}$ or $T_{n_1,m}$ can be composed by some 2RDFs of $CY_{n_2,m}$, where $n_1 > n_2$ (e.g. a 2RDF of $CY_{6,m}$ can be composed by a 2RDF of $CY_{2,m}$ and a 2RDF of $CY_{4,m}$). But the upper bounds obtained by this way are not tight (e.g. In table 1, UB of $CY_{2,4}$ is 4 and UB of $CY_{4,4}$ is 8, but γ_{r2}^P of $CY_{6,4}$ is 11).

We shall try to give a formal proof for lower bound in the future. Another interesting future work is that study of 2-rainbow domination on cylinders $CY_{n,m}$ and tori $T_{n,m}$ for $n \geq 6$.

References

- [1] M. Ali, M. T. Rahim, M. Zeb, and G. Ali, On 2-rainbow domination of some families of graphs, *International Journal of Mathematics and Soft Computing* 1 (2011) 47-53.
- [2] B. Brešar, M.A. Henning, D.F. Rall, Paired-domination of Cartesian products of graphs and rainbow domination, *Electronic Notes in Discrete Mathematics* 22 (2005) 233-237.
- [3] B. Brešar, M.A. Henning, D.F. Rall, Rainbow domination in graphs, *Taiwanese Journal of Mathematics* 12 (1) (2008) 213-225.
- [4] B. Brešar, T.K. Šumenjak, On the 2-rainbow domination in graphs, *Discrete Applied Mathematics* 155 (2007) 2394-2400.
- [5] G.J. Chang, J. Wu, X. Zhu, Rainbow domination on trees, *Discrete Applied Mathematics* 158 (2010) 8-12.
- [6] S. Fujita, M. Furuya, C. Magnant, k-Rainbow domatic numbers, *Discrete Applied Mathematics* 160 (2012) 1104-1113.
- [7] T.W. Haynes, S.T. Hedetniemi, P.J. Slater, *Fundamentals of Domination in Graphs*, Marcel Dekker, NewYork, 1998.
- [8] T.W. Haynes, S.T. Hedetniemi, P.J. Slater (Eds.), *Domination in Graphs: Advanced Topics*, Marcel Dekker, NewYork, 1998.
- [9] D. Meierling, S.M. Sheikholeslami, L. Volkmann, Nordhaus-Gaddum bounds on the k-rainbow domatic number of a graph *Applied Mathematics Letters*, Volume 24, Issue 10, October 2011, Pages 1758-1761
- [10] K.J. Pai, Y.C. Jhou, Upper Bounds on the 2-rainbow Domination Number of Grids, *Proceeding of the 29th Workshop on Combinatorial Mathematics and Computation Theory (CMCT 2012)*, April 27 - 28, Taipei, Taiwan, (2012) pp. 69-72.
- [11] T.K. Šumenjak, D.F. Rall, A. Tepeh, Rainbow domination in the lexicographic product of graphs, *Combinatorics* (2012), arXiv:1210.0514
- [12] C.L. Tong, X.H. Lin, Y.S. Yang, M.Q. Lou, 2-rainbow domination of generalized Petersen graphs $P(n, 2)$, *Discrete Applied Mathematics* 157 (2009) 1932-1937.
- [13] Y. Wu, N. J. Rad, Bounds on the 2-rainbow domination number of graphs, *Graphs and Combinatorics*, DOI 10.1007/s00373-012-1158-y
- [14] Y. Wu and H. Xing, Note on 2-rainbow domination and Roman domination in graphs, *Applied Math. Letter* 23 (2010) 706-709.
- [15] G. Xu, 2-rainbow domination of generalized Petersen graphs $P(n, 3)$, *Discrete Applied Mathematics* 157 (2009) 2570-2573.